

Outreach Agent: Complete System Design

Vision & Abstract

Build an intelligent, human-in-the-loop recruitment outreach pipeline that automates company research, resume customization, and cold mail drafting at scale (50+ companies/batch). The system acts as a research + personalization engine, not an autopilot spammer — every mail goes through HITL approval before sending, and follow-up workflows respect boundaries and response types. Success is measured by match quality (your fit 1-10) and conversion rates, not volume. Future iterations will A/B test templates and send timings to continuously optimize.

Core Problem Being Solved

Cold outreach at scale is time-intensive and often generic. This agent:

- Finds 50+ relevant hiring companies in 1 workflow run
 - Researches each company (funding, growth, sentiment, news)
 - Extracts hiring managers + validates email addresses
 - Customizes your resume without breaking the format
 - Drafts personalized cold mails based on company research + your fit
 - Tracks responses intelligently (no-response vs rejection vs interview request)
 - Triggers smart follow-ups (1 nudge after 7 days, but only if no response)
 - Logs everything to a master Google Sheet for analytics + prep
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Proposed Architecture

Layer 1: Discovery & Intelligence

Goal: Find 50 hiring companies + extract decision-maker emails + build company profiles

Role/Interest Prompt (natural language)

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Web Search Pipeline:

- Search for companies hiring in that space/seniority
- Extract job board links, company careers pages

- Identify relevant roles + hiring signals

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Company Intelligence Gathering:

- Funding: Recent Series A/B/C announcements?
- Growth: Headcount trends, hiring velocity (roles posted/week)
- News: Recent launches, layoffs, product announcements (with timestamps)
- Sentiment: Glassdoor/blend scores, public perception
- Leadership: Founder backgrounds, investor profiles
- Traffic: Similarweb or Crunchbase metrics

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Email Extraction:

- LinkedIn scraping: HR, TA, hiring managers (with titles, profiles)
- Company careers page parsing
- Email validation (syntax check, bounce detection where possible)

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Store in Google Sheet: Company profile row

Output: List of 50 companies with structured data (name, funding round, growth signal, leadership, sentiment, email contacts)

Layer 2: Matching & Quality Gate

Goal: Score your fit for each company/role, flag weak matches for review

For each Company + Role:

- Parse JD for: skills, tech stack, seniority level, responsibilities
- Cross-reference your resume + projects against JD keywords
- Scoring algorithm:

- * Tech stack match (0-3 points)
- * Seniority alignment (0-2 points)
- * Project relevance (0-3 points)
- * Company growth/signal match (0-2 points)

= Total: 1-10 score

- Threshold: If score < 6, flag for manual review before proceeding

(Prevents spamming weak matches)

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Store in Google Sheet: Match score, keyword gaps, reasoning

Output: Prioritized list of high-match companies ($\geq 6/10$) ready for personalization

Layer 3: Resume Customization

Goal: Tailor resume to each company without breaking format/layout

For each High-Match Company:

- Extract JD keywords missing from your current resume
- Identify which section of resume can be edited:
 - * Skills section: Add relevant keywords
 - * Experience bullets: Reword to emphasize relevant impact
 - * Projects section: Highlight project that aligns with JD
- LaTeX Editing Strategy:
 - * Parse resume LaTeX (using regex/pylatexenc)
 - * Identify editable blocks (bullets, skill tags, project descriptions)
 - * Rewrite content to emphasize job-relevant achievements

- * Preserve: section headers, formatting, fonts, spacing
- * Keep: Overall structure (Education, Experience, Projects, Skills)

- PDF Generation:

- * Compile LaTeX → PDF using pdflatex/latexmk

- * Validate: Is it exactly 1 page?

- * If > 1 page: Auto-condense

- Shorten bullet points

- Reduce margins slightly (within reason)

- Remove less relevant projects

- Regenerate + revalidate

- * If still > 1 page: Flag for manual review (shouldn't happen)

- Store PDF with company name: resume_CompanyName.pdf

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Store in Google Sheet: Customization summary (which keywords added, which section edited)

Output: Customized, single-page PDF resume for each company

Layer 4: Cold Mail Drafting

Goal: Write personalized, conversational cold mails that show you've researched the company

For each High-Match Company:

- Research to include:

- * Recent company milestone (funding, product launch, news)

- * Role context (what they're hiring for, why it matters)

* Your fit (connect your projects/skills to their needs)

* Your value add (what specific problem you'd solve)

- Mail Template Structure (researched format):

Greeting:

"Hi [Hiring Manager Name]," (personalized, not "Dear Hiring Team")

Hook:

1-2 sentences: Why you're reaching out to *this* company specifically

(Show you researched them: "I saw you just closed Series B and are scaling the data platform team...")

Body (Conversational, not formal):

Para 1: Brief intro (who you are, current role/student status)

Para 2: Your relevant experience (2-3 bullets connecting to their JD)

Para 3: Why you're interested in them specifically (company mission, tech stack, growth stage)

Para 4: What you'd contribute (specific value prop for this role)

CTA:

"Open to a quick chat about the role—happy to hop on a call whenever works best."

OR

"Would love to learn more about the team. Let me know if my background is a fit!"

Sign-off:

Your name, LinkedIn URL, GitHub link, phone number

- Tone Guidelines:

- ✓ Natural, conversational (like you're talking to a peer)
- ✓ Show research (mention company detail, recent news, product)
- ✓ Specific to role (not generic "I'm interested in your company")
- ✗ No corporate jargon
- ✗ No overselling or flowery language
- ✗ Not too long (8-10 sentences max)

- Attachment:

Customized resume PDF (named: Prajwal_Pandey_[CompanyName].pdf)

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Store Draft in Gmail (NOT SENT YET)

Add: [HR/TA Email] to "To", [Your Email] to "From"

Subject: "[Role Name] Application - Prajwal Pandey"

Output: Gmail draft (HITL approval required before sending)

Layer 5: Google Sheets Master Database

Goal: Single source of truth for all outreach data, response tracking, prep

Master Sheet Schema:

Company & Role Data:

- Company Name
- Role Title
- Company URL / LinkedIn
- Job Board Link (where role was found)
- Funding Status (Series X, amount, date)

- Growth Signal (headcount YoY, hiring velocity)
- Recent News (funding/launch/layoff, timestamp)
- Sentiment Score (Glassdoor, Blind, public perception)
- Founder/Leadership names + backgrounds

Contact Data:

- Hiring Manager Name
- Hiring Manager Title
- HR/TA Email
- Email Validation Status (valid/bounce/unknown)
- LinkedIn Profile URL

Your Fit & Research:

- Match Score (1-10)
- Keyword Gaps (what was missing from resume)
- Resume Customizations Applied
- Customized Resume PDF Link (Google Drive)

Mail & Tracking:

- Mail Draft Date
- Mail Sent Date (only if HITL approved + sent)
- Mail Status (draft, sent, bounced)
- Recipient Response Type:
 - * No response (X days)
 - * Polite rejection
 - * "We're not hiring"

- * Auto-reply detected
- * Forwarded to recruiter
- * Interview request

- Response Date
- Response Details (note from email)

Follow-up Logic:

- Follow-up Due Date (sent date + 7 days)
- Follow-up Sent? (Y/N)
- Follow-up Response Type
- Follow-up Status (closed, waiting, interview scheduled)

Interview Prep (Only if Positive Response):

- Company Profile Deep-Dive:
 - * Size, locations, valuation
 - * Employee count, growth rate
 - * Public sentiment, recent news
 - * Top products, market position
- Role Deep-Dive:
 - * Day-to-day responsibilities
 - * Tech stack details
 - * People in similar roles (names, backgrounds from LinkedIn)
 - * Team size, reporting structure
 - * Sentiment about role (Blind, Glassdoor, internal reviews)

- Your Fit & Prep:

- * Why you fit this company/role
- * 3 key areas to prepare
- * Projects to highlight in interviews
- * Common questions they ask (from Glassdoor, blind)

- Interview Structure:

- * Number of rounds
- * Round names (screening, coding, design, system design, etc)
- * Format (online, in-person, async)
- * What they ask in each round
- * Estimated prep time
- * Top 10 questions they typically ask

Metrics & Analysis:

- Total companies researched
- Companies passed quality gate ($\geq 6/10$)
- Mails sent (HITL approved)
- Response rate (%)
- Interview request rate (%)
- Average time to response
- Best-performing keywords/skills

Layer 6: Response Tracking & Follow-Up Orchestration

Goal: Automate smart follow-ups, respect boundaries, avoid spam

Response Classification (Manual + Automated):

Automated Detection:

- Auto-reply detection: "I'm out of office", "auto-generated"
- Forwarded to recruiter: Email comes from different sender
- Interview request: Email contains "interview", "call", "next steps"

Manual Classification (for ambiguous emails):

- User flags as: no response, rejection, "not hiring", forwarded, or interview request

Follow-up Workflow (Only if "No Response"):

Day 0: Mail sent (HITL approved)

Day 7: Check for response

- If response exists: Log response type, stop
- If no response: Send 1 follow-up nudge email

Subject: "Following up on [Role] at [Company]"

Content: Brief, natural reminder (2-3 sentences max)

Attach: Same customized resume

Create as Gmail draft (HITL approval again)

Day 14+: If still no response → Mark as "no response after 14 days", close loop

No Follow-up If:

- Explicit rejection received
- "We're not hiring" response

- Interview request (move to prep)
- 3+ follow-ups already sent (avoid spam)

Scheduling Rules:

- Mails only sent Mon-Fri, 8:30 AM (India Standard Time)
- Follow-ups respect same scheduling rule
- Timezone: IST (your location)
- Future A/B test: different send times, different templates

Output: Automated reminders for follow-ups; all drafts in Gmail awaiting HITL approval

Layer 7: Interview Prep Deep-Dive (Triggered on Positive Response)

Goal: Detailed company/role/interview research after you get a response

Triggered When: Interview request OR forwarded to recruiter

Company Deep-Dive Research:

- Web search: Company news, recent announcements, funding rounds
- Extract: Size, locations, employee count, valuation, growth rate
- Leadership: Founder bios, backgrounds, investor profiles
- Products: What they build, market position, recent launches
- Culture: Glassdoor score, Blind reviews, employee sentiment
- Hiring: Recent hiring announcements, team expansion
- Red flags: Layoffs, scandals, turnover rates
- Timeline: When was funding announced? When did they hire last?

Role Deep-Dive Research:

- Job description deep analysis (if available)
- Similar roles at company: What do people in this role do?
- Tech stack: Languages, frameworks, tools they use
- Team structure: Who's the manager? Team size?
- Career progression: Where do people move after this role?
- Sentiment: Blind/Glassdoor posts about this role
- Growth potential: Do people get promoted from this role?

Your Fit Analysis:

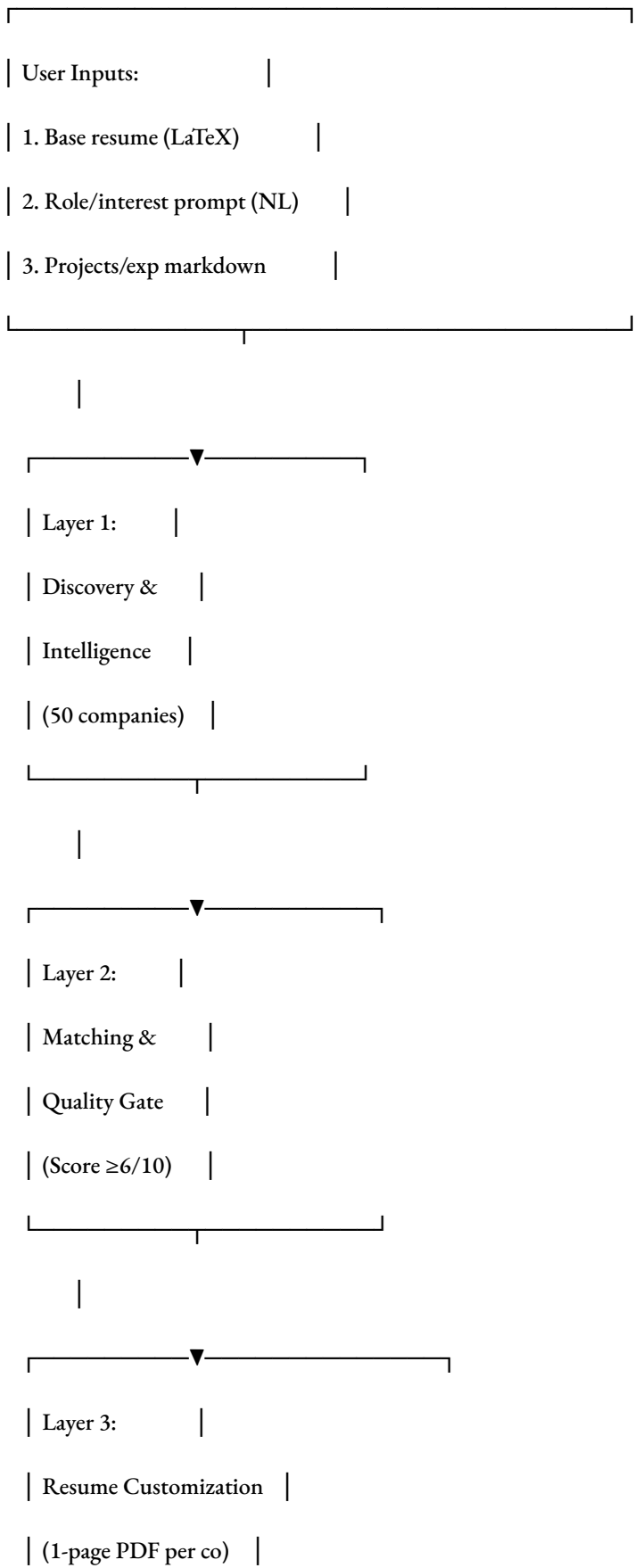
- Why you fit this company (mission, tech, growth stage)
- Why you fit this role (skills, projects, interests)
- What to emphasize in interviews (your strongest selling points)
- Gaps to address (if any, and how to talk about them)
- Questions to ask them (showing research + interest)

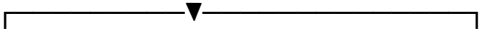
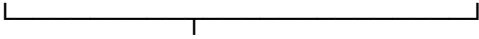
Interview Structure & Prep:

- Research: Glassdoor, Blind, Leetcode company forums
- Extract: Number of rounds, format, what they ask
- Scoring: Common questions, patterns in their interviews
- Prep plan: What to study, mock interview topics
- Timeline: How long are interviews? How far apart?

Output: Add all this to Google Sheet (interview prep tab or section)

Complete Data Flow

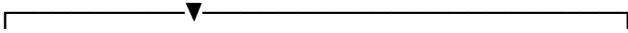
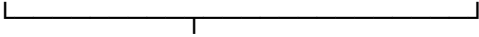




| Layer 4: |

| Cold Mail Drafting |

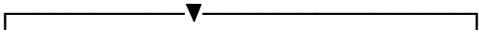
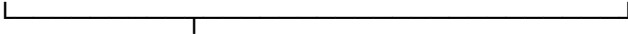
| (Gmail draft, HITL) |



| Layer 5: |

| Google Sheet Master DB |

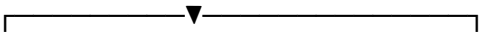
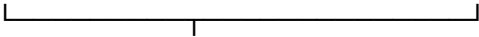
| (All data logged) |



| USER DECISION POINT: |

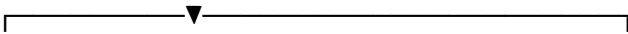
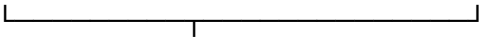
| Review + Approve Mails |

| (HITL gate) |



| Send Approved Mails |

| (Mon-Fri, 8:30 AM IST) |

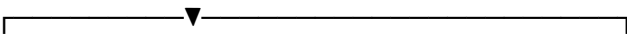
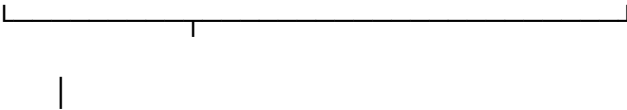


| Layer 6: |

| Response Tracking & |

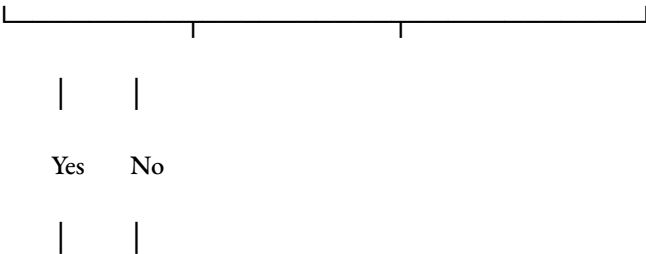
| Follow-up Orchestration |

| (Day 7 nudge if no response) |



| Positive Response Received? |

| (Interview request or forwarded) |

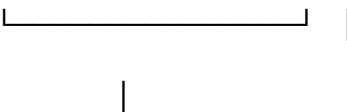


| Layer 7: | |

| Interview | |

| Prep Deep- | |

| Dive Research | |



| Close Loop |

| (Mark as complete) |



Tech Stack

Backend/Core

- Python 3.10+: Main orchestration language
- LangGraph / LangChain: Agent orchestration + workflow management
- Claude API (via Anthropic SDK): LLM for research, drafting, analysis
- Web Search: Claude's built-in web search tool (via API)

Data & Storage

- Google Sheets API ([gspread](#) Python lib): Master database for all outreach data
- Google Drive API: Store customized resume PDFs
- Gmail API ([google-api-python-client](#)): Create/manage draft emails, schedule sends

Resume Processing

- `pylatexenc` / `regex`: Parse + edit LaTeX resume
- `pdflatex` / `latexmk` (system call): Compile LaTeX to PDF
- `PyPDF2` / `pdfplumber`: Validate page count, extract metadata

Email & Communication

- Gmail API: Drafting, tracking, scheduling (not SMTP—use Gmail's native draft system)
- `email-validator`: Validate extracted email addresses
- `python-cron` / `APScheduler`: Schedule follow-ups at 8:30 AM Mon-Fri IST

Email Extraction (Optional, can be done manually/semi-manually)

- BeautifulSoup / Scrapy: Scrape company careers pages, LinkedIn (if automating)
- RapidAPI: Email finder APIs (Hunter.io, Clearbit) if going automated

Testing & Monitoring

- `pytest`: Unit tests for LaTeX parsing, PDF validation, email formatting
- logging: Detailed logs for debugging agent decisions
- Sentry / custom logger: Track agent errors in production

Infrastructure (Deployment, Future)

- Docker: Containerize the agent
 - GitHub Actions / Airflow: Schedule daily/weekly runs
 - PostgreSQL (optional): Backup/version control for Sheet data
 - Streamlit / FastAPI: Dashboard to review drafts, approve sends (UI layer)
-

Detailed System Flow (Step-by-Step)

Phase 1: Initialization (User-Provided)

1. User provides:

- base_resume.tex
- projects_and_exp.md
- role_interest_prompt: "AI/ML engineer roles at early-stage startups with <500 people"

2. Agent validates inputs:

- Resume: Parse LaTeX, check for sections (Experience, Skills, Projects)
- Projects: Parse markdown, extract key technologies
- Prompt: Analyze for hiring signals (seniority, company stage, tech)

Phase 2: Company Discovery (Automated)

1. Web Search Pipeline:

- Query 1: "AI/ML engineer roles hiring startup <500 employees 2026"
- Query 2: "Series A/B funding announcements AI startups India 2026"
- Query 3: "ML engineer job openings early-stage tech startups"
- Parse results: Extract company names, job board links, careers pages

2. For each company found (50 total):

- Deep web search: Company site, LinkedIn, Crunchbase, news
- Extract: Funding history, recent announcements, leadership
- Score: Is this relevant to your interest? (1-10 scoring)
- Filter: Keep only $\geq 6/10$ matches

3. Email extraction:

- Search: LinkedIn hiring manager profiles
- Parse: Company careers page for contact info
- Extract: HR, TA, hiring manager emails
- Validate: Check email format, optional bounce check

4. Log to Google Sheet: Company row with all metadata

Phase 3: Resume Customization (Per Company)

For each high-match company:

1. Analyze JD:

- Extract: Required skills, tech stack, keywords
- Compare: Against your resume
- Identify: Missing keywords, underemphasized skills

2. Edit LaTeX:

- Strategy: Preserve structure, nudge content
- Find editable blocks: Skill bullets, experience descriptions, projects
- Rewrite: Emphasize job-relevant achievements
- Example:

Before: "Built a real-time data pipeline"

After: "Built a real-time data pipeline in Apache Kafka and Python

processing 100k+ events/sec for ML model inference"

(Added specifics to match JD if it mentioned "Kafka" + "real-time")

3. Compile & Validate:

- Run pdflatex: resume_temp.tex → resume_temp.pdf
- Check: Page count == 1?
- If > 1 page:
 - Condense: Shorten bullets, reduce whitespace
 - Regenerate: Recompile PDF
 - Recheck: Page count
- If still > 1: Flag for manual review

4. Store:

- Save PDF: /resumes/CompanyName_resume.pdf
- Upload to Google Drive (for sharing later)
- Log to Sheet: Customizations applied

Phase 4: Cold Mail Drafting (Per Company)

For each customized resume:

1. Gather research snippets:

- Recent funding: "Acme AI raised Series B (\$50M) last month"
- Product milestone: "Just launched their new product X"
- Role context: "Hiring for ML engineers to scale their recommendation engine"
- Your connection: "Your background in recommendation systems aligns perfectly"

2. Draft mail structure:

Greeting: "Hi [Hiring Manager Name],"

Hook: Reference recent company milestone + reason for reaching out

Body: Your background (2-3 relevant bullets) → Why this role → Why them

CTA: "Open to chatting more about the role!"

Sign-off: Name, LinkedIn, GitHub, phone

3. Tone check:

- Read aloud: Does it sound like a human talking to a peer?
- Remove: Any corporate jargon or overselling
- Add: Specific company details (show research)
- Length: 150-200 words max

4. Create Gmail Draft:

- To: hiring_manager_email@company.com
- Subject: "[Role Name] Application - Prajwal Pandey"
- Body: Cold mail text
- Attachment: resume_CompanyName.pdf
- Status: DRAFT (not sent)

5. Log to Sheet:

- Draft date, hiring manager name, email, subject, status

Phase 5: HITL Approval Gate

User reviews all 50 drafts:

- Read mail: Does it make sense? Is it personalized?
- Check resume: Does it look right? Any formatting issues?
- Decision: Approve to send OR edit/reject

Approved drafts → Ready to send

Rejected drafts → Agent can revise or user manually edits

(This is your manual safety check before emails leave your account)

Phase 6: Sending (Scheduled)

For each approved mail:

1. Check schedule:

- Is it Mon-Fri, 8:30 AM IST? If not, queue for next available slot
- Spread sends: Don't send all 50 at once (looks spammy)
- Batch: Send 5-10 per day across the week

2. Send mail:

- Gmail API: Send draft → convert to sent
- Log: Sent date/time

3. Update Sheet:

- Status: "Sent"
- Sent Date: ISO timestamp
- Follow-up Due: Sent Date + 7 days

Phase 7: Response Tracking

Background job (runs daily or on-demand):

1. Check inbox for replies:

- Filter: Only emails from contacts in Sheet
- Classify response:
 - * Auto-reply: "I'm out of office" → Auto-flag
 - * Interview request: "Let's schedule a call" → Auto-flag
 - * Forwarded: Email from different sender → Auto-flag
 - * Rejection: "Not a fit" / "Decided to go with other candidates" → Manual review
 - * No response: Silence → Flag for 7-day follow-up

2. Log response:

- Response Type: Classification above
- Response Date: Date received
- Response Details: Copy relevant text from email
- Update Sheet

3. Trigger actions:

- Interview request → Start Layer 7 (prep research)
- No response + Day 7 passed → Queue follow-up email
- Rejection → Close loop (no follow-up)
- Auto-reply detected → Check again in 5 days

Phase 8: Follow-Up Workflow (If No Response)

Day 7 after send:

1. Check: Did they respond?

- Yes → Log response, done

- No → Proceed with follow-up

2. Draft follow-up:

- Subject: "Following up on [Role] at [Company]"
- Body: 2-3 sentences, brief reminder + CTA
- Tone: Casual, not pushy
- Attachment: Same customized resume
- Create as Gmail draft (HITL approval again)

3. User review + approve (same as Phase 5)

4. Send (if approved):

- Same scheduling rules (Mon-Fri, 8:30 AM)
- Log follow-up sent date

5. Track again:

- If response: Log it, done
- If no response after 14 days: Mark as "no response", close loop

Phase 9: Interview Prep (Triggered on Positive Response)

When: Interview request OR forwarded to recruiter

1. Company Deep-Dive:

- Search: Company recent news, funding, products, leadership
- Extract: Size, locations, valuation, growth, sentiment
- Research: Team, hiring trends, culture signals

- Compile: 2-3 page summary in Sheet

2. Role Deep-Dive:

- Search: Similar roles at this company, day-to-day tasks
- Extract: Tech stack, team structure, progression
- Research: Blind posts about role, sentiment
- Compile: Role overview, growth potential, questions they ask

3. Your Fit Analysis:

- Connect: Your projects ↔ Company needs
- Highlight: 3 key strengths for this role
- Prepare: Common questions, how to answer
- Compile: Personalized prep guide

4. Interview Structure:

- Search: Glassdoor, Blind, LeetCode company forums
- Extract: Number of rounds, format, topics
- Compile: "Expected Rounds" section in Sheet
- Add: Top 10 questions they typically ask

5. Prep Plan:

- Create: Week-by-week study guide
- Suggest: Mock interview topics
- Link: Resources (LeetCode problems, system design guides)

All logged to Sheet under "Interview Prep" section

Google Sheets Master Database Schema (Detailed)

Column Structure (One Row = One Company + Role)

Identification

- A: Company Name
- B: Role Title
- C: Company URL
- D: Job Board Link

Company Intelligence

- E: Funding Status (e.g., "Series B - \$50M, June 2026")
- F: Company Size (Headcount estimate)
- G: Locations (HQ + offices)
- H: Growth Signal (Hiring velocity, YoY headcount growth)
- I: Recent News (Latest announcement + timestamp)
- J: Sentiment Score (Glassdoor/Blind aggregate, 1-5)
- K: Founders/Leadership (Names + brief bios)
- L: Public Valuation / Revenue (if available)

Contacts

- M: Hiring Manager Name
- N: Hiring Manager Title
- O: HR/TA Email
- P: Email Validation Status (Valid / Bounce / Unknown)
- Q: LinkedIn Profile URL (Hiring Manager)

Your Fit

- R: Match Score (1-10)
- S: Keyword Gaps (e.g., "Missing: Kubernetes, Docker")
- T: Resume Customizations Applied (Bullet point list)
- U: Customized Resume Link (Google Drive shareable URL)

Mail Tracking

- V: Draft Date (ISO timestamp)
- W: Mail Status (Draft / Sent / Bounced / No Response / Positive Response)
- X: Sent Date (ISO timestamp, only if sent)
- Y: Response Type (No response / Rejection / Interview request / Auto-reply / Forwarded / Not hiring)

- Z: Response Date (Date email received)
- AA: Response Details (Copy of response email text)

Follow-Up Tracking

- AB: Follow-up Due Date (Sent Date + 7 days)
- AC: Follow-up Status (Not Yet / Sent / Response Received / Closed)
- AD: Follow-up Sent Date (If sent)
- AE: Follow-up Response Type

Interview Prep (Only if positive response)

- AF: Company Deep-Dive Summary (Rich text link to detailed notes)
- AG: Role Deep-Dive Summary
- AH: Your Fit Analysis
- AI: Interview Structure (# of rounds, types)
- AJ: Top 10 Interview Questions
- AK: Prep Plan (Link to study guide)
- AL: Interview Schedule (Dates if scheduled)

Notes & Tags

- AM: Notes (Internal observations, red flags, etc)
- AN: Tags (e.g., "high-priority", "startup", "well-funded", "needs-follow-up")

Technology Details & Justification

Why Claude API + LangGraph?

- Claude: Best for research, personalization, nuanced writing (cold mails need to sound human)
- LangGraph: Agent workflows need state management, branching logic, tool calls → LangGraph handles this elegantly

Why Google Sheets (not SQL)?

- You need a master DB, but also human-readable interface
- Sheets = Single source of truth + easy to pivot/filter manually
- Backup: Can export to CSV anytime
- Future: If scale grows, migrate to PostgreSQL + Sheets for sync

Why Gmail API (not SMTP)?

- Gmail drafts = built-in HITL gate (sits in your inbox awaiting review)
- Easier to track opens, replies (Gmail thread context)
- No need to manage sending infrastructure

Why LaTeX (not DOCX)?

- You specified Jakes Overleaf format (LaTeX is version-controllable, precision formatting)
- PDF output = universal, one-page guarantee is trivial to validate

Why 8:30 AM Mon-Fri?

- Data: Cold mails sent during work hours have higher open rates
 - Future A/B test: Try different times (9 AM, 6 PM?) to find your optimal window
-

Future Potential Features

Phase 2 (Next-Level Intelligence)

- Investor Tracking: Map investors → their portfolio companies → recent funding = hiring signal
- Hiring Momentum Scoring: Track companies posting new roles daily (hiring fast = more likely to respond)
- Keyword Bidding: Track which keywords get you most interviews, auto-weight them in resume customization
- Response Time Patterns: Some companies reply in 24 hours, others in 2 weeks—log these patterns

Phase 3 (A/B Testing)

- Template Variants: Draft 3 versions of each mail (personal touch, data-driven, humble), A/B send 50/50
- Timing Experiments: Send some mails at 8:30 AM, some at 6 PM, measure response rates
- Subject Line Testing: "Following up + question" vs "Quick Question" — track opens/responses
- Resume Tweaks: Test different resume formats (ATS-optimized vs design-heavy), measure callbacks

Phase 4 (Automation Extensions)

- Auto-screening calls: Schedule video call links in follow-up mails (Calendly integration)
- Portfolio builder: Auto-generate tailored portfolio PDF for each company (projects you built + company relevance)
- Offer letter analyzer: When offer comes, parse + compare with other offers (salary, equity, title)
- Slack/Discord bot: Notify you of responses in real-time (don't need to check email constantly)

Phase 5 (ML + Prediction)

- Response predictor: After 30+ mails, train model: What company/role attributes → highest response rate?
- Interview prep predictor: Based on company data, predict what they'll ask (fine-tune on Blind/Glassdoor data)
- Offer probability: Given your match score, estimate likelihood of getting an offer

Phase 6 (End-to-End Automation)

- Interview scheduling: When interview request arrives, auto-check your calendar + suggest times
 - Interview notes: AI transcribes your interviews, highlights key questions + your answers
 - Offer negotiation: Draft counter-offer based on market data, competitor offers, your value
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Success Metrics

Short-term (1 month):

- companies researched: 50+
- high-match companies ($\geq 6/10$): 40+
- mails sent (after HITL approval): 30-40
- positive responses: 5-10 (goal: 15-25% response rate)

Medium-term (3 months):

- interview requests: 3-5
- offers: 1+
- Response rate trending (which companies/industries respond best?)
- Optimal send times / subject lines (from A/B tests)

Long-term (6+ months):

- ML model accuracy: Can you predict who'll respond?
 - Benchmark: Your personalized approach vs. generic mass mail (do 10 personalized > 50 generic?)
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Deployment & Scaling

MVP (Single-Run, Manual)

- Python script + Jupyter notebook
- User runs: `python outreach_agent.py --role_prompt "..." --resume resume.tex`
- Output: Gmail drafts + Google Sheet populated
- User reviews & approves all mails before sending

v1.0 (Scheduled, Semi-Automated)

- Wrap in FastAPI or Streamlit for UI
- GitHub Actions: Run discovery + research nightly
- User dashboard: Review pending drafts, approve sends
- Gmail API: Auto-check responses, classify, trigger follow-ups

v2.0 (Full Automation, Self-Improving)

- Scheduled agent runs weekly
 - A/B testing framework built-in
 - ML models: Response predictor, offer likelihood, interview prep
 - Slack notifications: "New response from X company", "Follow-up due", "Interview in 3 days"
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Risk Mitigation

Email Getting Marked as Spam

- Mitigation: Personalize heavily (generic = spam), don't send >20 on same day, use legit sender (your Gmail)
- Monitor: Check spam folder weekly, adjust send volume if needed

Getting Blacklisted

- Mitigation: Only send to extracted email addresses (not company domains), one follow-up only (not spammy)
- Validate: Email validation before sending

Resume PDF Generation Fails

- Mitigation: Fallback to manual review if LaTeX compile fails, don't auto-send broken PDFs
- Test: Always validate page count before Gmail draft creation

Hiring Manager Doesn't Exist

- Mitigation: LinkedIn validation (check if profile is real before extracting email)
- Fallback: Send to general HR email if hiring manager lookup fails

Low Response Rate

- Mitigation: Quality gate (don't send <6/10 matches), heavy personalization, A/B test timing + templates
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Summary

You're building a research → personalization → drafting → tracking → follow-up engine for cold outreach at scale. The key insight is: 10 hyper-personalized mails beat 50 generic ones. The system enforces this through match scoring, resume customization, and company research before you ever draft a mail.

HITL gates ensure you never spam accidentally. Google Sheets gives you a master database for analytics + future optimization. A/B testing framework future-proofs you for continuous improvement.

The agent does the hard work (research, customization, drafting) in hours instead of weeks. You do the human work (reviewing, deciding, tweaking) before anything goes out.

That's your moat: *quality + volume + data.